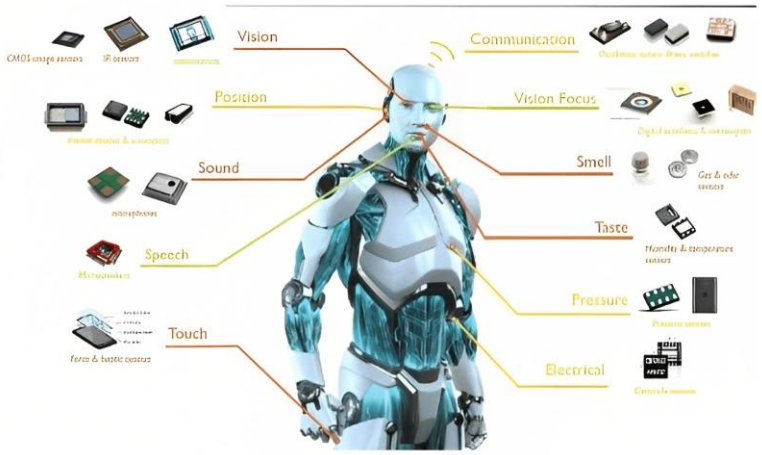
RO2 - Lecture 2
Perception in Mobile Robotics

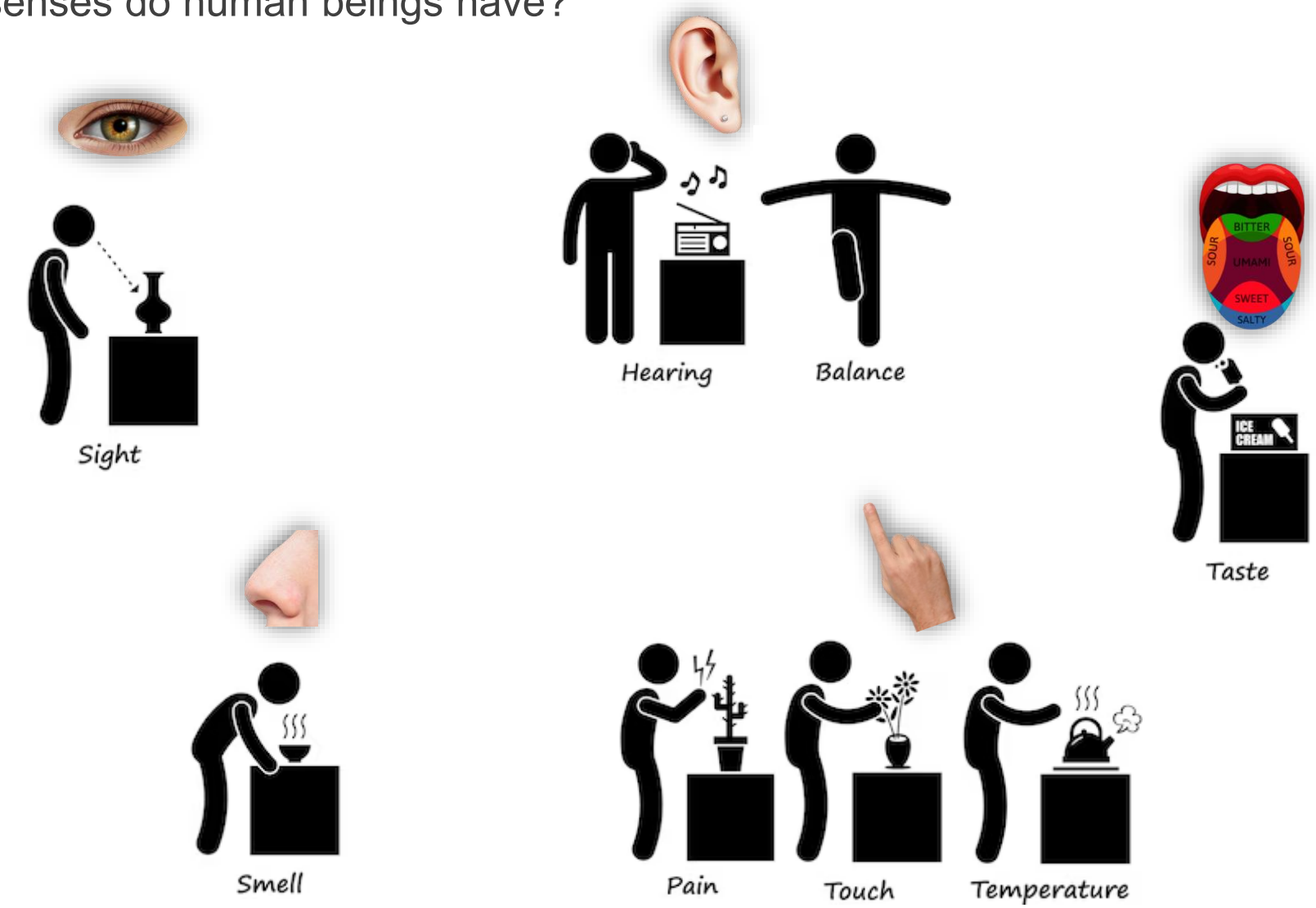


we focus on students

Fachhochschule Dortmund
University of Applied Sciences and Arts

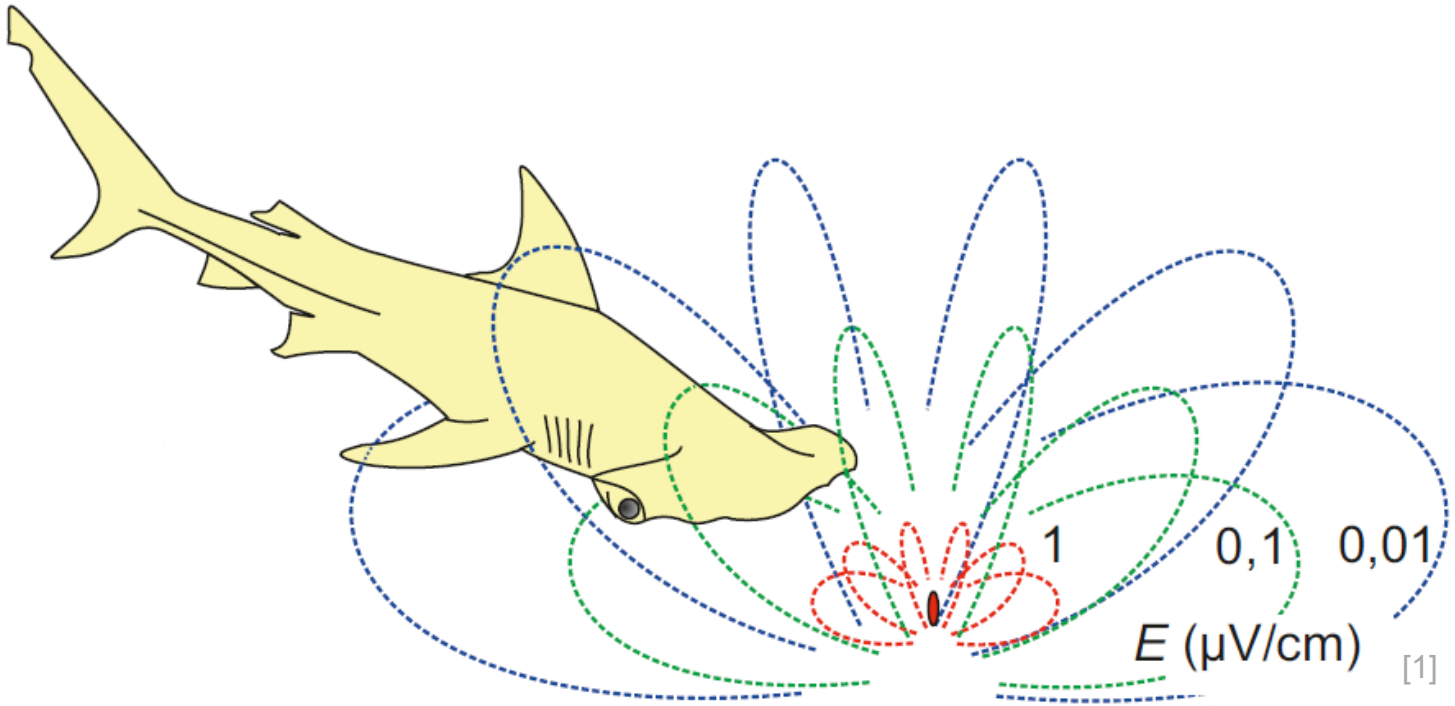
1 Perception in Mobile Robotics

- Which senses do human beings have?



- Which other senses do animals have (examples)?

Hammerhead
Electroreceptive Sense

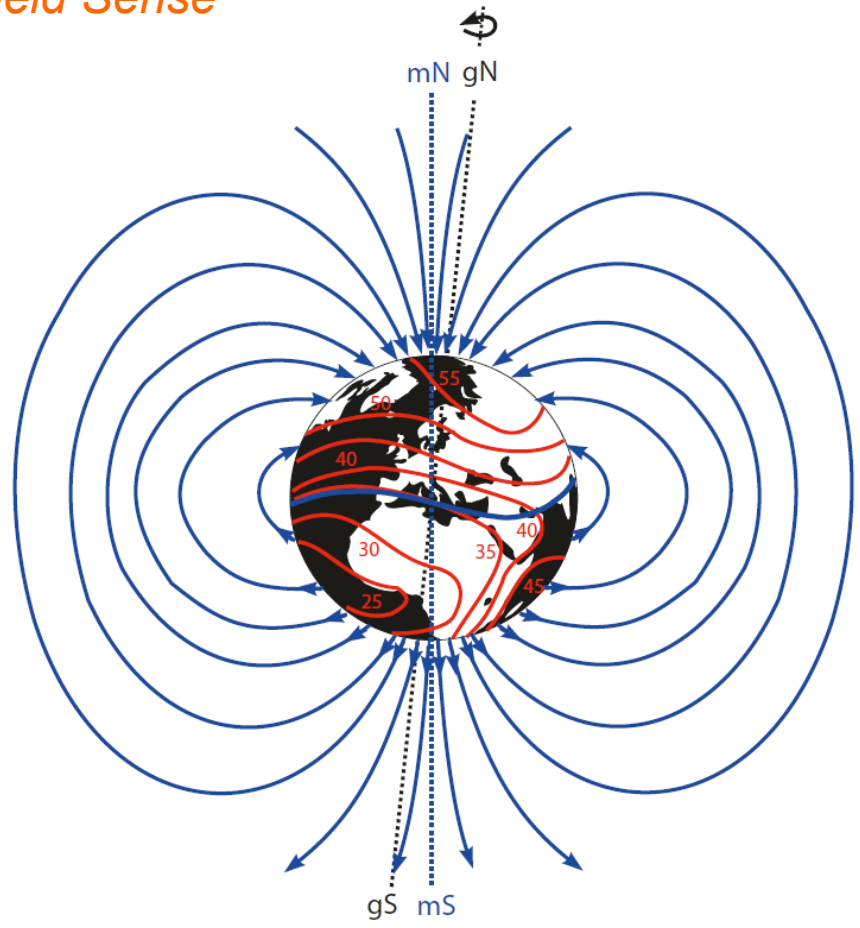


[1]

- Which other senses do animals have (examples)?

Migratory Birds

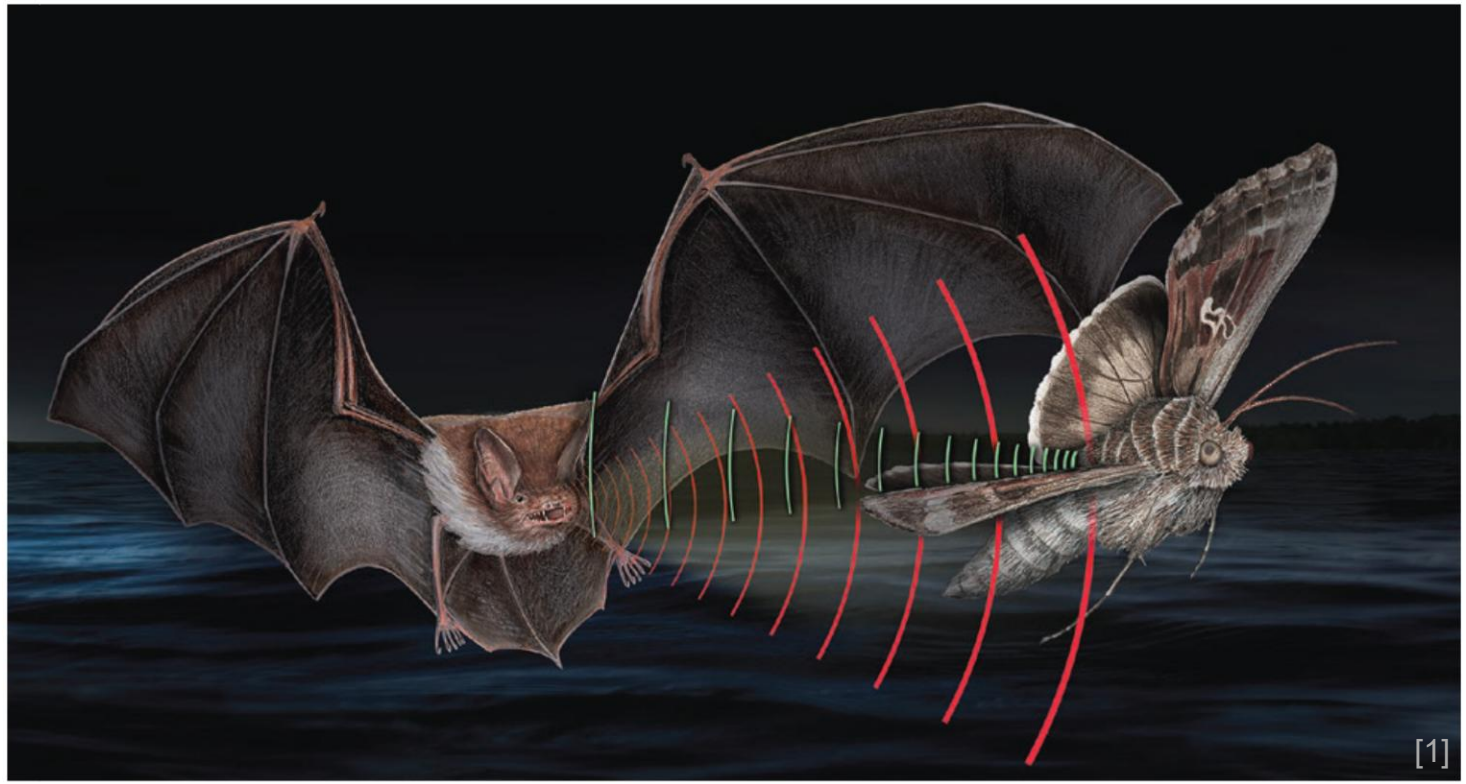
Magnetic Field Sense



- Which other senses do animals have (examples)?

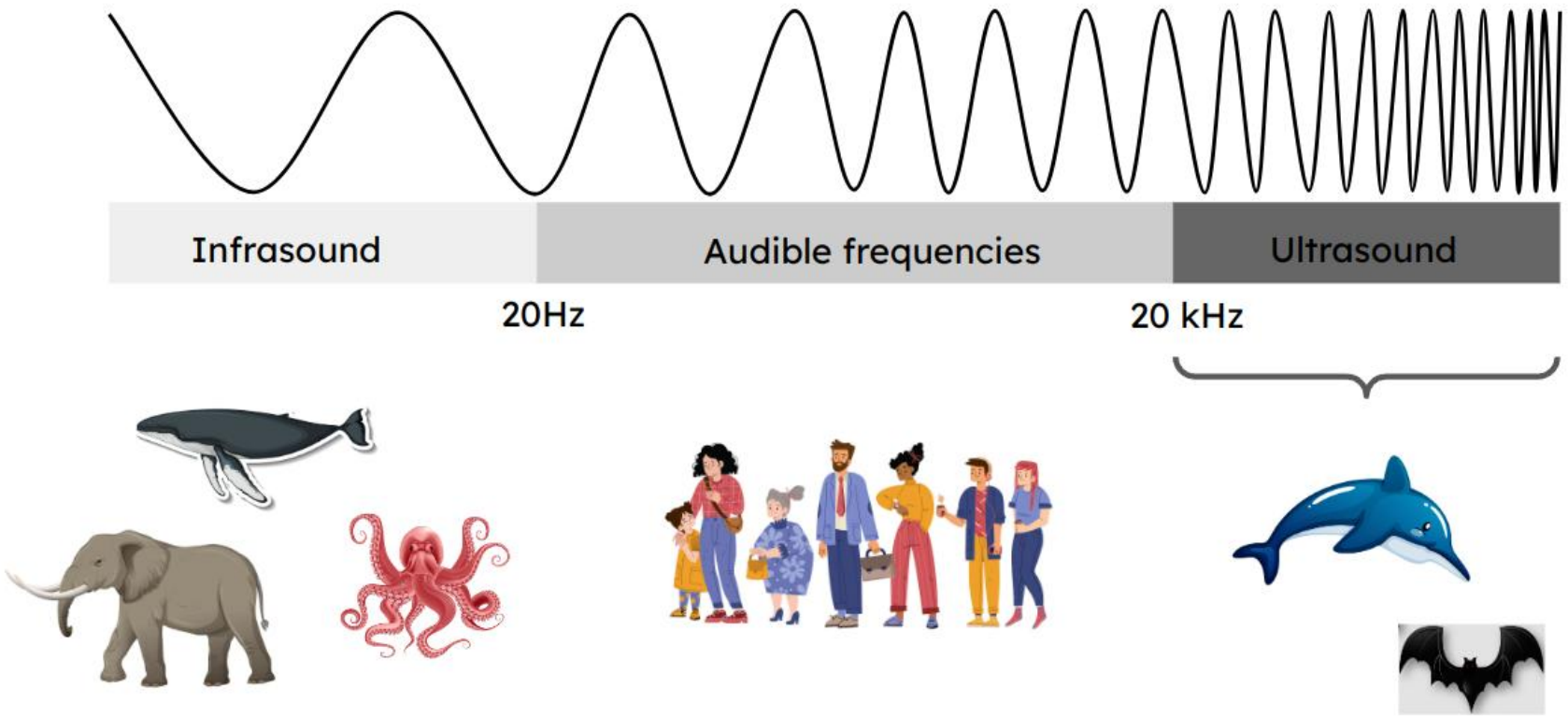
Bats

Sense of hearing (ultrasonic waves)



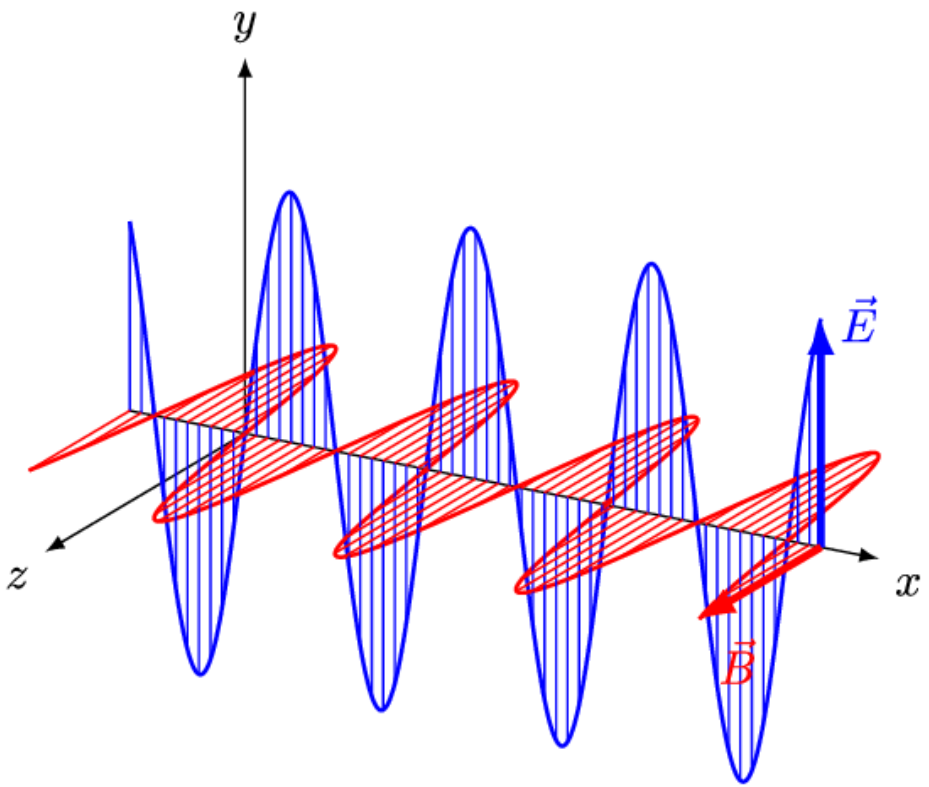
- Which other senses do animals have (examples)?

Sense of hearing (ranges)



Sensing of electromagnetic waves – Basics

- Electromagnetic waves are oscillations of *electric fields* ***and*** *magnetic fields* that propagate through space
- The waves oscillate perpendicular to each other and perpendicular to their direction of movement (transverse waves)
- Do not require a medium: can travel through vacuum as well as through various materials
- All electromagnetic waves travel with the speed of light:
  - in vacuum: 299,792,458 m/s
  - in air: 299,702,547 m/s (refractive index = 1.0003)
  - In water: 225,407,863 m/s (refractive index = 1.33)



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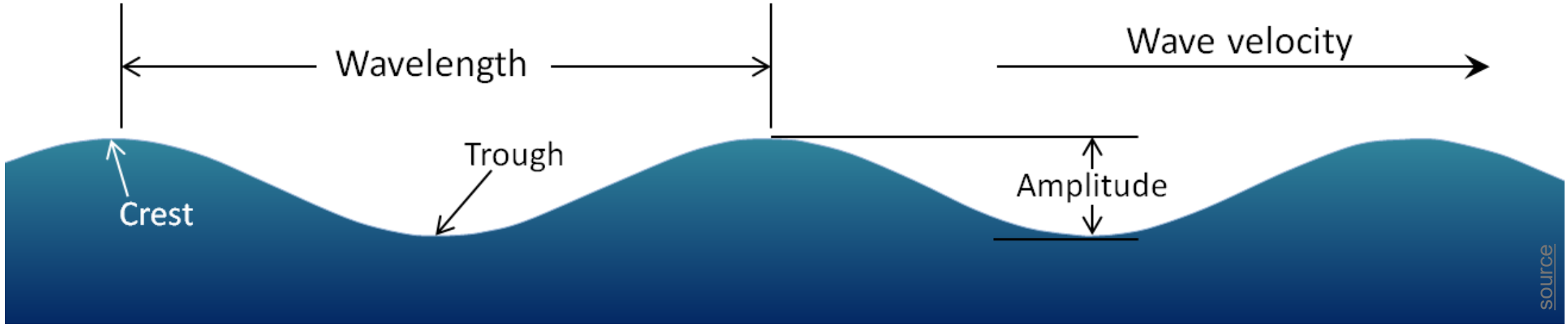
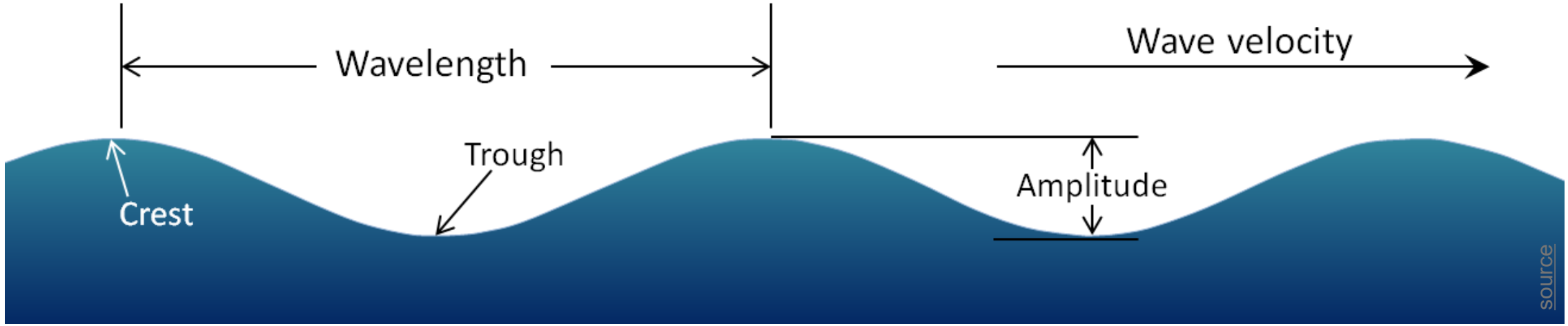
Sensing of electromagnetic waves – Wavelength and Fequency



- **Wave period** is the **duration** of time it takes for one complete cycle or oscillation of a wave to pass a fixed point (measured in seconds)
- **Frequency** is the number of cycles or oscillations of a wave that occur in a unit of time (measured in Hertz (Hz)), indicating how often the wave repeats itself per second.
- The **Wavelength** of a wave is the **spatial period** of a periodic wave - the **distance** over which the wave's shape repeats (measured in meters)

Sensing of electromagnetic waves – Wavelength and Fequency

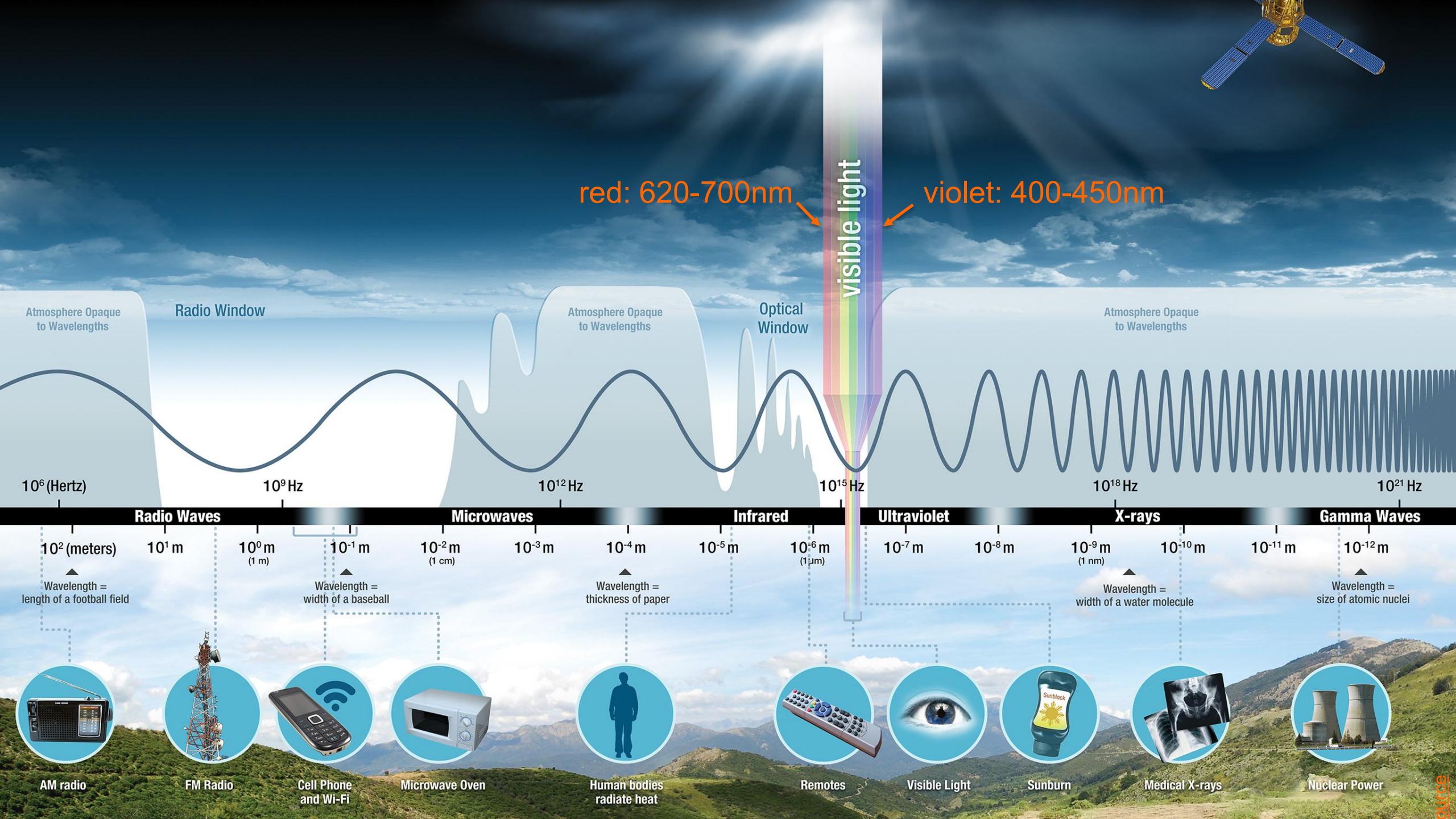


- Relation between Wavelength λ and Wave frequency f:

$$\lambda = \frac{v}{f}$$

, where v is the propagation speed (speed of light)

- The wavelength λ is inversely proportional to the wave frequency f ($v = \text{const}$)


red: 620-700nm
violet: 400-450nm
visible light
Atmosphere Opaque to Wavelengths
Radio Window
Atmosphere Opaque to Wavelengths
Optical Window
Atmosphere Opaque to Wavelengths
10^6 (Hertz)
10^9 Hz
10^{12} Hz
10^{15} Hz
10^{18} Hz
10^{21} Hz
Radio Waves
Microwaves
Infrared
Ultraviolet
X-rays
Gamma Waves
10^2 (meters)
10^1 m
10^0 m (1 m)
10^{-1} m
10^{-2} m (1 cm)
10^{-3} m
10^{-4} m
10^{-5} m
10^{-6} m (1 μm)
10^{-7} m
10^{-8} m
10^{-9} m (1 nm)
10^{-10} m
10^{-11} m
10^{-12} m
Wavelength = length of a football field
Wavelength = width of a baseball
Wavelength = thickness of paper
Wavelength = width of a water molecule
Wavelength = size of atomic nuclei
AM radio
FM Radio
Cell Phone and Wi-Fi
Microwave Oven
Human bodies radiate heat
Remotes
Visible Light
Sunburn
Medical X-rays
Nuclear Power

Sensing of electromagnetic waves – Main attributes based on wavelength/frequency

Energy E that can be carried by an electromagnetic wave:

$$E = h \cdot f$$

where h is the Planck's constant (a constant relating the frequency of the wave to the energy of the photon)

Electromagnetic waves with higher frequencies (shorter wavelengths) carry more energy, while waves with lower frequencies (longer wavelengths) carry less energy.

Animal examples for perception of electromagnetic waves other than light

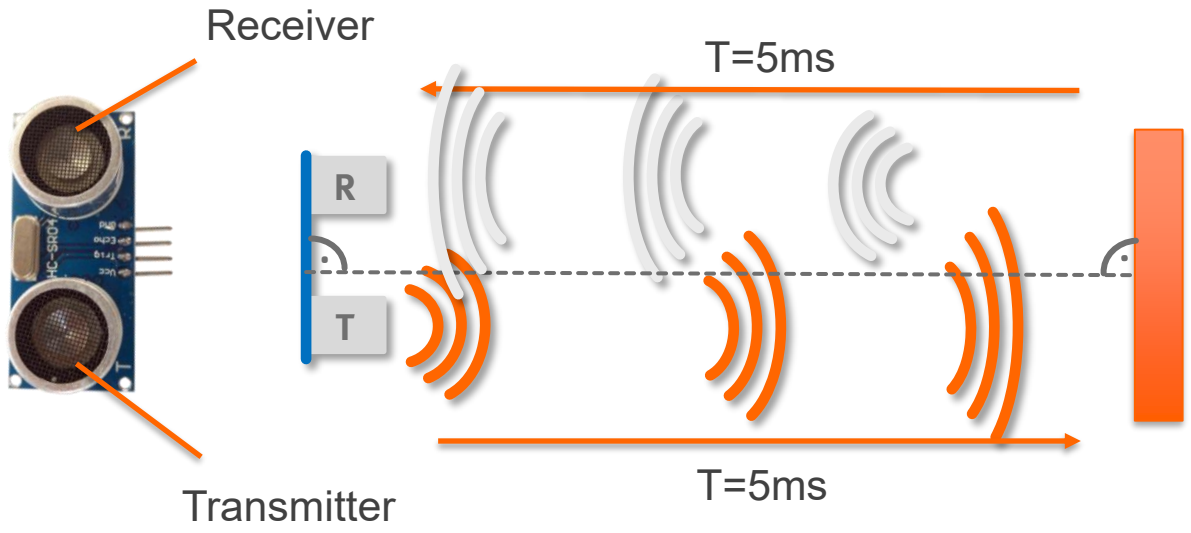
Bees use **UV light** to find nectar in flowers

Reindeers use **UV light** to spotting predators and food in the snow

Rattlesnake sense **infrared** radiation, enabling them to sense the body heat of prey

Melanophila can detect **infrared** radiation from forest fires, which they use to find freshly burned wood for laying eggs

Ultrasonic based distance estimation



- Ultrasonic sensors emit **high-frequency sound waves** (20-40 kHz)
- The sensor measures the **time it takes** for the sound wave to bounce back from an object (echo) using the equation:

$$Distance = \frac{Time\ of\ Flight}{2} \cdot Speed\ of\ Sound$$

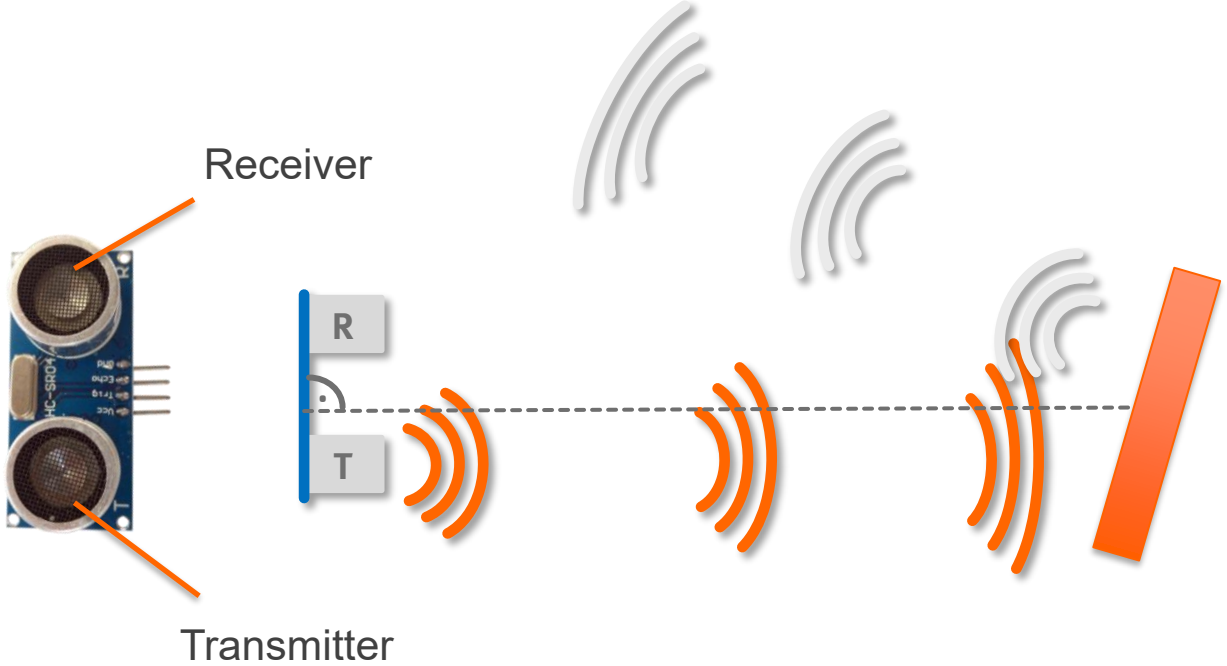
- Speed of sound (air) approximately:

$$c_A \approx \left(331{,}5 + 0{,}6\frac{\vartheta}{°C}\right)\frac{m}{s}$$

- E.g. ToF = 10ms, Temperature ϑ = 20°C

$$Distance = \frac{10}{2} \cdot 10^{-3}s \cdot 343{,}5\frac{m}{s} = 1{,}72m$$

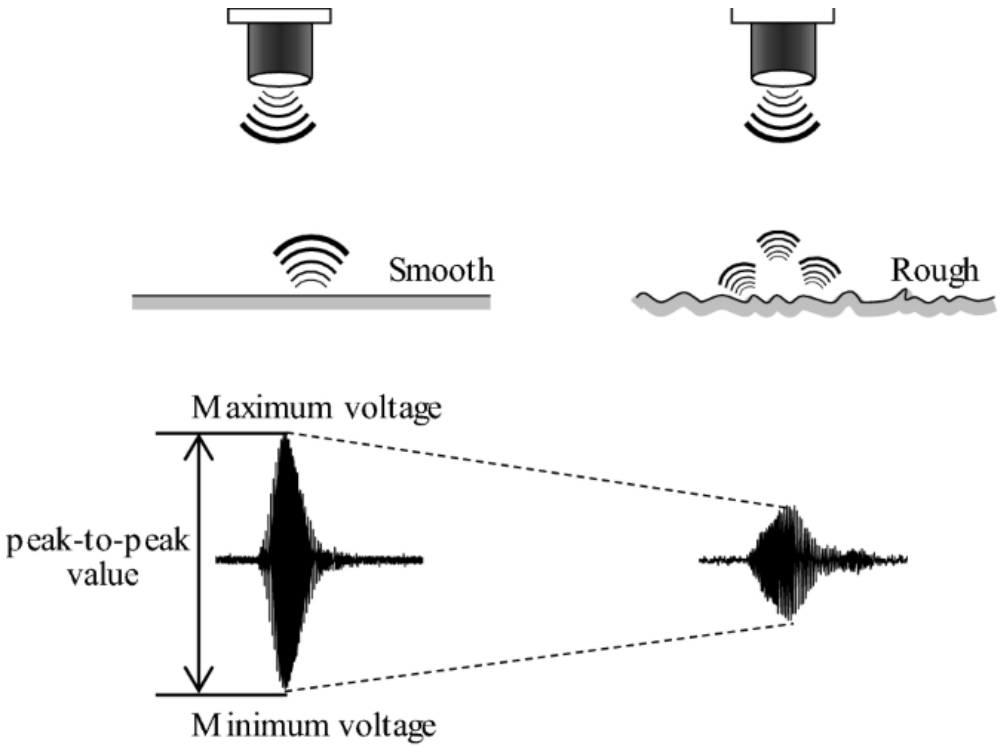
Ultrasonic based distance estimation - limitations



Angle of Incidence:

If the ultrasonic waves hit a smooth, flat surface at an angle -> it does not return directly to the sensor

Ultrasonic based distance estimation - limitations



Aerial ultrasonic reflection wave [2]

Surface Texture:

- The texture of the surface being measured can significantly affect how sound waves are reflected back to the sensor
- Smooth, hard surfaces reflect sound waves well, while rough or irregular surfaces scatter the sound, leading to less accurate measurements

Ultrasonic based distance estimation - limitations

Humidity and Temperature:

- speed of sound in air changes with temperature. If the temperature is not taken into account, the distance calculated based on a default speed of sound (like 343 m/s) can lead to errors
- Humidity affects the density of air (humid air is less dense than dry air) → influence on speed of sound (higher humidity can increase the speed of sound slightly)

Ultrasonic based distance estimation - limitations

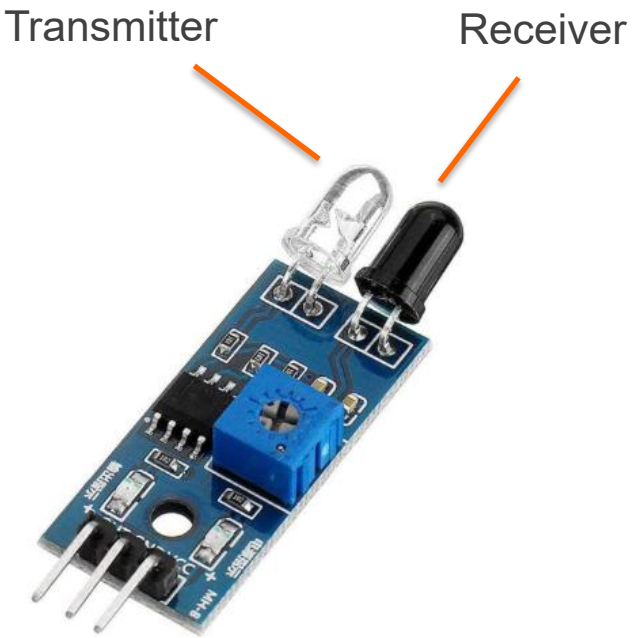
Absorbing Materials:

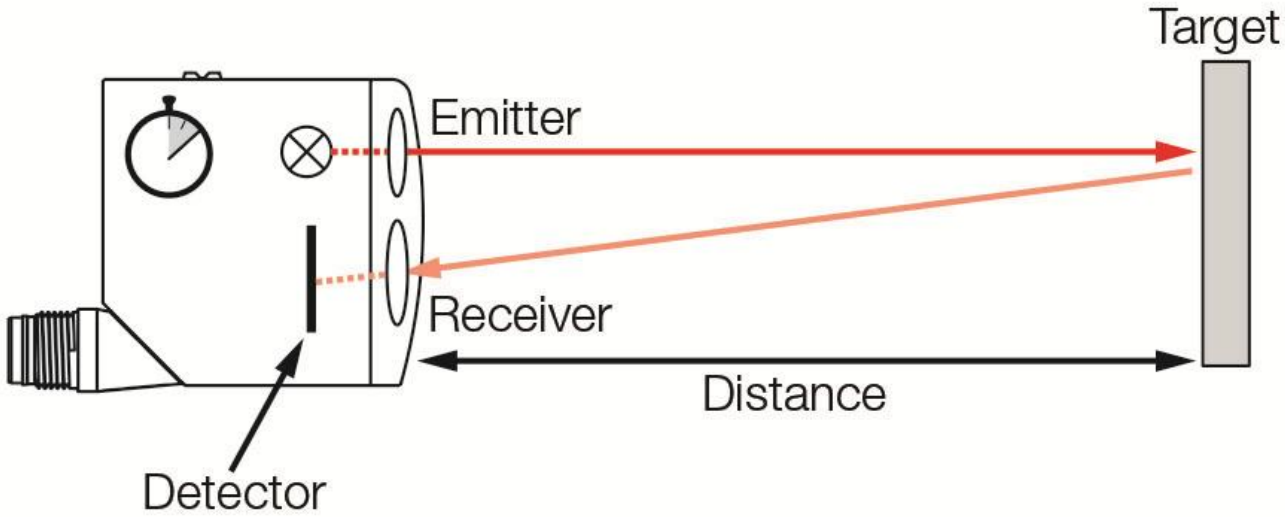
- Certain materials absorb ultrasonic waves, leading to weaker echoes resulting in unreliable distance readings
- E.g.: foam, fabric, or carpet absorb sound waves, causing a significant reduction in the intensity of the echo that returns to the sensor

Ultrasonic Sensors – Pros/Cons

Advantages	Disadvantages
• Versatile for indoor and outdoor applications	• Performance drops if the target is angled
• Non-contact measurement reduces wear and tear	• Ineffective on sound-absorbent materials
• Measures distances from few centimeters to 10 meters	• Affected by temperature, humidity
• Easy integration with microcontrollers	• Less accurate over longer distances
• Affordable	• Prone to ultrasonic and noise interference
• Durable for industrial use	• Higher power consumption compared to some alternatives

Sensing of electromagnetic waves – Infrared Sensors





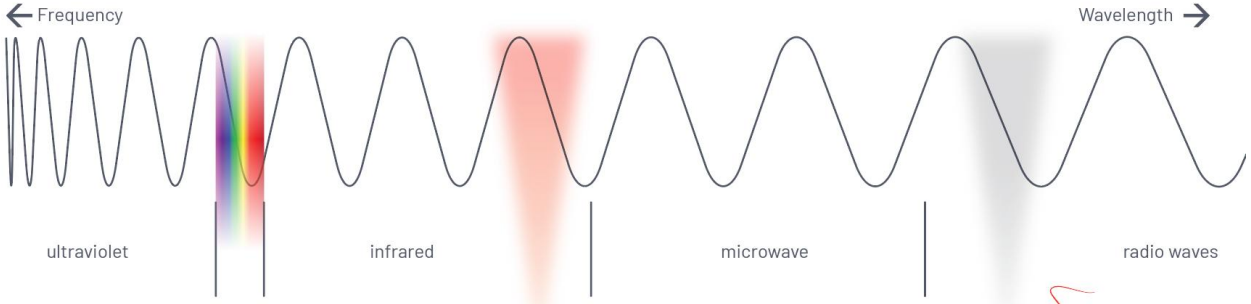
IR sensors are typically used for proximity detection

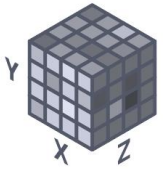
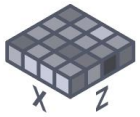
- **IR emitter** (e.g., an LED) sends out infrared light, **receiver** (e.g., a photodiode) detects the reflected light from nearby objects
- wavelength range is typically between 700 nm to 1 mm

Infrared Sensors – Pros/Cons

Advantages	Disadvantages
• **Fast response time**	• **Sensitivity to light environment** (e.g. bright sunlight can reduce accuracy)
• **Compact and leightweight** (e.g. smaller form factor compared to ultrasonic sensors)	• IR sensors usually have a **shorter detection range** (few centimeters to a few meters) compared to ultrasonic sensors
• **Low power consumption** (also suited for battery driven devices)	• Reduced performance for objects with when detecting transparent or highly absorptive materials
• **Low cost** (e.g. cheaper than ultrasonic sensors)	• distance measurements are **less precise** compared to ultrasonic sensors

Sensing of electromagnetic waves – Radar, LiDAR, Camera



	CAMERA COLOR	LIDAR 3D RANGE	RADAR AXIAL SPEED
Wavelengths	~380nm (Violet) to ~780nm (Red)	840nm/905nm (Close to visible) 1550nm (Eye safest)	4mm (77Ghz) Higher resolution 12mm (24Ghz) (First version)
Spatial coverage			
Strong points	High Resolution images	Native 3D data	Robustness to weather conditions

source

Radar based estimation for distance and environmental perception

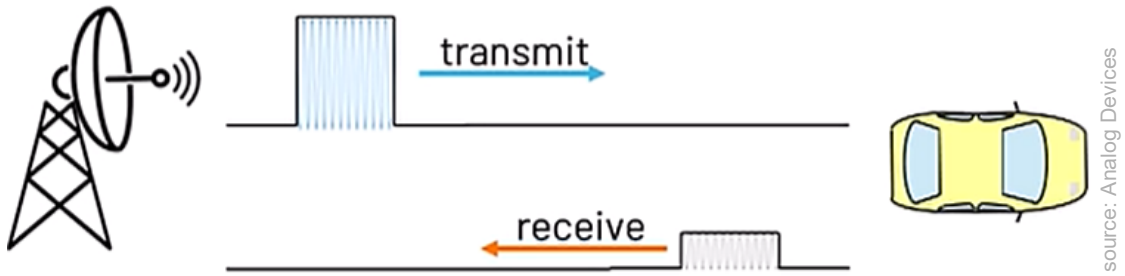
- **Radar**: ***Ra**dio **D**etection **a**nd **R**anging*
- Radar systems transmit electromagnetic waves (1mm…1m) and receive the waves reflected back from objects (targets)
- the properties of the received echo (time delay, frequency shift, intensity) provide information about the target
- Single transmitter/receiver → estimate **range** and **velocity**
- Scanning (mechanical/electronic) or the use of multiple receivers allows **angle estimation**
- With multiple transmitter antennas and multiple receiver antennas (MIMO systems), a radar can estimate the **2D angle detection** (azimuth and elevation) and improved resolution

source: giphy

Radar based estimation for distance and environmental perception

Common measurement techniques:

- **Pulsed Radar:** discrete pulses are sent and the time delay for the echoes is measured

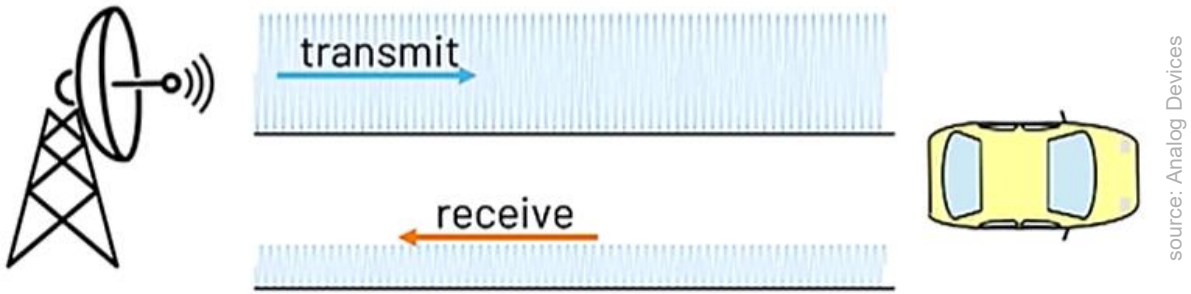


- Can measure *distance* to an object based on the pulse travel time, but not the speed of an object (at least with a single pulsed signal)

Radar based estimation for distance and environmental perception

Common measurement techniques:

- **Continuous Wave (CW) Radar:** a *continuous sinusoidal signal with fixed frequency* is transmitted and the difference of the frequency in the received signal is determined (doppler effect)



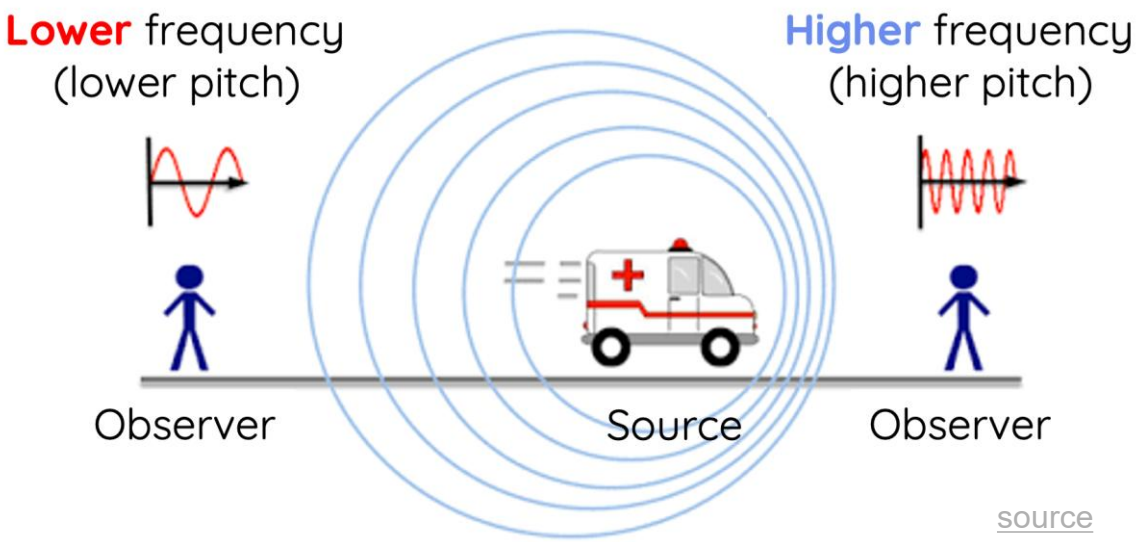
- Continuous wave emission provides *speed* measurement (based on doppler effect), but no distance measurement

Radar based estimation for distance and environmental perception

Refresher: Doppler Effect

- the Doppler Effect refers to the change in frequency (or wavelength) of a wave as perceived by an observer due to the relative motion of the source and/or the observe

- **Example sound of ambulance**:
  - when a sound source like the ambulance moves toward an observer, the sound waves are compressed, causing a higher frequency and thus a higher tone
  - when the source moves away from the observer, the sound waves are stretched out, leading to a lower frequency and a lower tone



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Radar based estimation for distance and environmental perception

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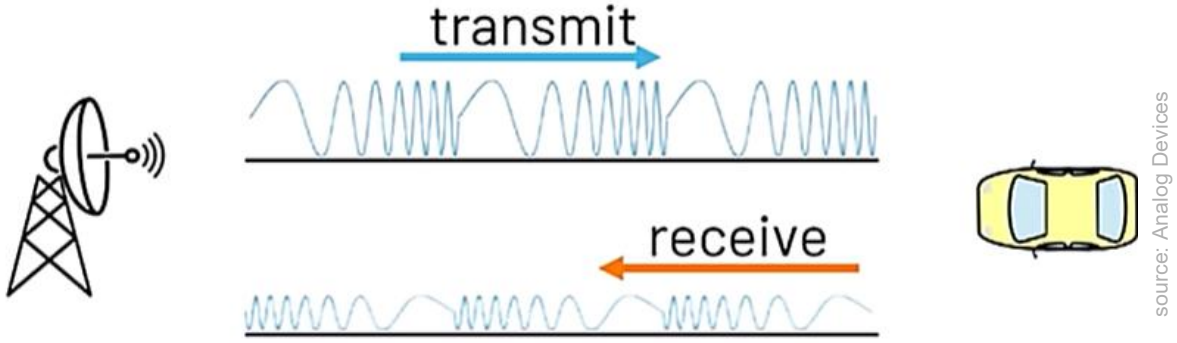
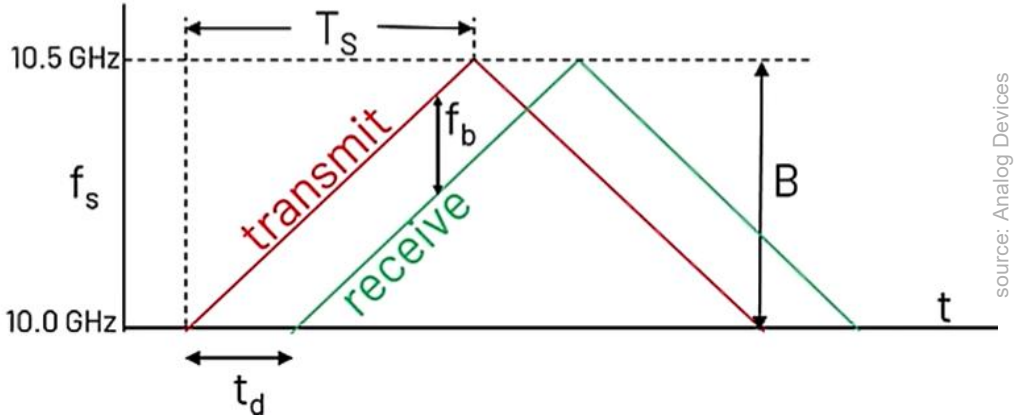
Radar based estimation for distance and environmental perception

Common measurement techniques:

- **FMCW Radar (Frequency Modulated Continuous Wave):** the transmitted signal is a sinusoidal signal in which *the frequency of the signal varies with time*



source: Analog Devices



source: Analog Devices

- Can reliably measure both *distance and speed*

Radar based estimation for distance and environmental perception

Pros and Cons:

- **Advantages**
  - Not affected by environmental effects: radar operates effectively in rain, fog, snow, or low visibility environments
  - Radar operates independently of ambient light, making it suitable for day and night use
  - Radar can accurately measure the speed and position of objects at long ranges, especially in dynamic environments
  - Radar systems are compact and relatively low cost (e.g. compared to lidar)

Radar based estimation for distance and environmental perception

Pros and Cons:

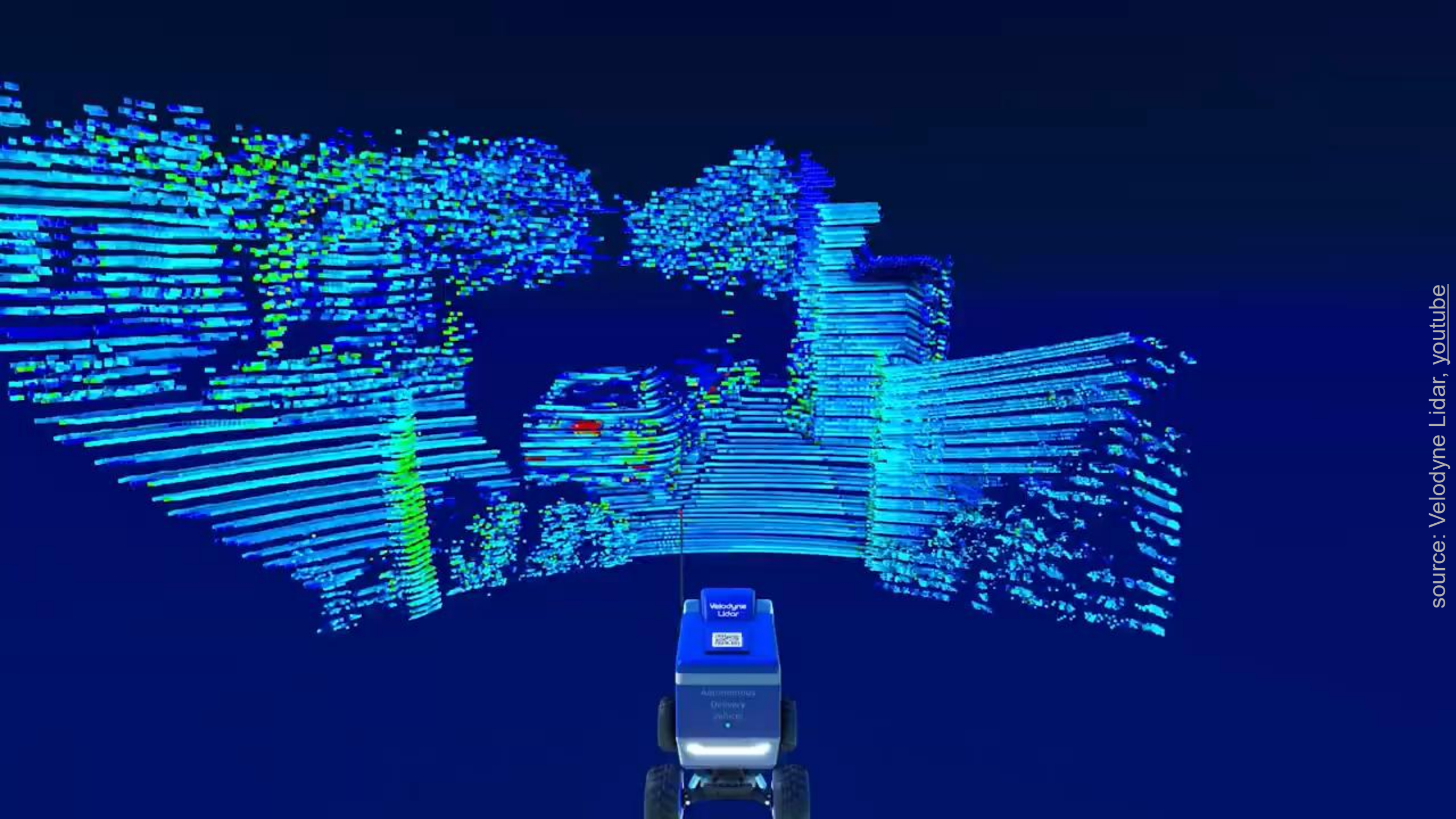
- **Disadvantages**
  - Radar typically provides less resolution than other technologies like lidar (less details)
  - Sensitivity to interference: Radar can be affected by other electronic devices emitting electromagnetic signals, causing noise or false readings
  - Radar data requires more advanced processing algorithms

LIDAR based estimation for distance and environmental perception

LiDAR: ***L**ight **D**etection **a**nd **R**anging*

- LiDAR systems illuminate a target with laser light and measure the time it takes for the pulses to return → calculation of distances based on the time of flight (ToF)
- laser in LiDAR refers to a focused beam of coherent light, typically infrared (e.g., 905 nm, 1550 nm) or visible light wavelengths

source: Pheonix LiDAR Systems, youtube



source: Velodyne Lidar, youtube

LIDAR based estimation for distance and environmental perception

Pros and Cons:

- **Advantages**
  - LiDAR provides high resolution data and detailed 3D point clouds, enabling accurate mapping and object detection
  - LiDAR thus offers precise object differentiation and distinguishing between closely spaced objects
  - LiDAR operates effectively in both day and night conditions, as it uses its own laser light source

LIDAR based estimation for distance and environmental perception

Pros and Cons:

- **Disadvantages**
  - Sensible to environmental effects: challenging weather as rain, fog, and snow can reduce LiDAR accuracy and reliability
  - LiDAR systems are very expensive compared to other sensing technologies (>10k €)
  - The effective range of lidar is often shorter than radar, especially for detecting distant objects
  - multiple lidar systems can interfere with one another, especially if they operate on similar wavelengths and emit pulses simultaneously

Camera based estimation for distance and environmental perception

- Cameras use *passive sensing* to interpret the world by capturing light from the environment (unlike radar and lidar that emit signals actively)
- Cameras generally capture 2D images that represent the visual surroundings includes details such as *color*, *texture*, *shapes*, and *movements*
- To obtain 3D information such as distance, different techniques are applied:
  - Depth estimation with a single camera is e.g. achieved using machine learning approaches where neural networks are trained to estimate depth from a 2D data

1 Perception in Mobile Robotics

source

Camera based estimation for distance and environmental perception

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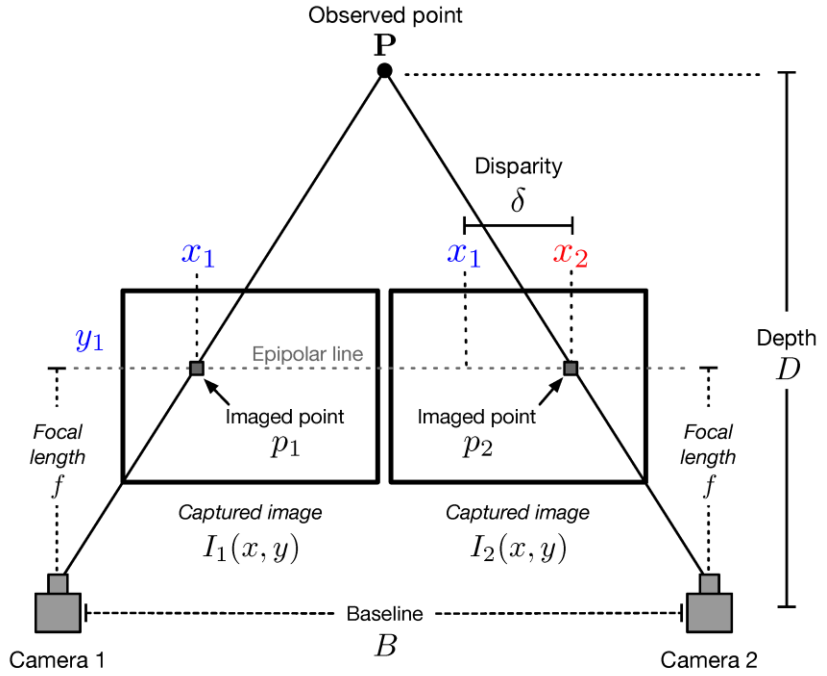
  - Depth estimation with a single camera is e.g. achieved using machine learning approaches where neural networks are trained to estimate depth from a 2D data
  - Stereo Vision (Two Cameras): by using two cameras spaced at a known distance apart (like human eyes), stereo vision can estimate the distance to objects by
    - comparing the two images and finding corresponding points (features) e.g. related to objects
    - calculating the difference in position of the object between the two cameras by triangulation



source

Camera based estimation for distance and environmental perception

Pros and Cons:

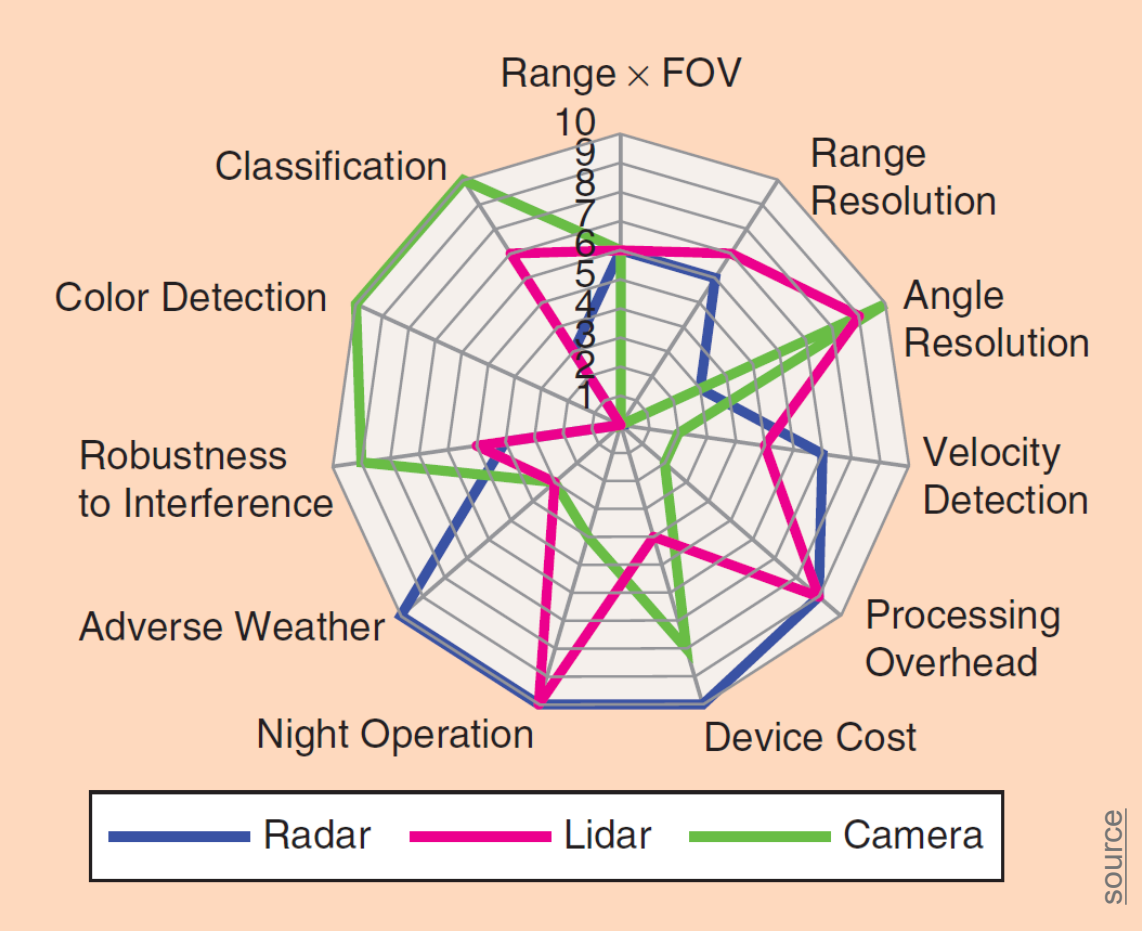
- **Advantages**
  - Camera pictures provide rich information like color, texture, and shapes, which is very helpful for object recognition (e.g., traffic signs, pedestrians etc.)
  - Very cost-effective compared to radar or lidar
  - Compact and lightweight, easy to integrate into vehicles or robots (flexible placement)
  - Low interference under “normal” conditions (e.g. with regard to weather/light)

Camera based estimation for distance and environmental perception

Pros and Cons:

- **Disdvantages**
  - Performance is highly affected by weather conditions such as rain, fog, snow, and dirt on the lens can obscure the view and reduce performance
  - Performance is highly affected by light conditions such as low light, glare, shadows, and overexposure
  - limited in estimating velocity and range (as depth, range, and motion are not directly measured)

Comparison: Radar, LiDAR, Camera



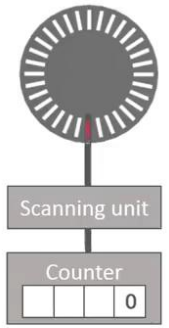
Odometry

- Odometry is a technique used by robots and vehicles to estimate their position in space over time by tracking the number of revolutions made by the wheels, which can be converted into linear and rotational motion
- The number of revolutions is determined using *rotatory encoders*
- For example, if a robot has two wheels and both spin simultaneously forward → the travelled distance can be determined based on the number of wheel rotations
- Angular Displacement can be determined from the difference in rotation between two wheels, providing orientation changes (yaw)

Odometry

Incremental Encoders

- Incremental Encoders use discs which contain gaps that are equally spaced
- An LED emits a pulsed light through the gaps in the rotating disc which is detected by a photo detector
- The position and the rotational speed can be determined based on the measured pulses per revolution



Schematics of an incremental measuring system with an encoder

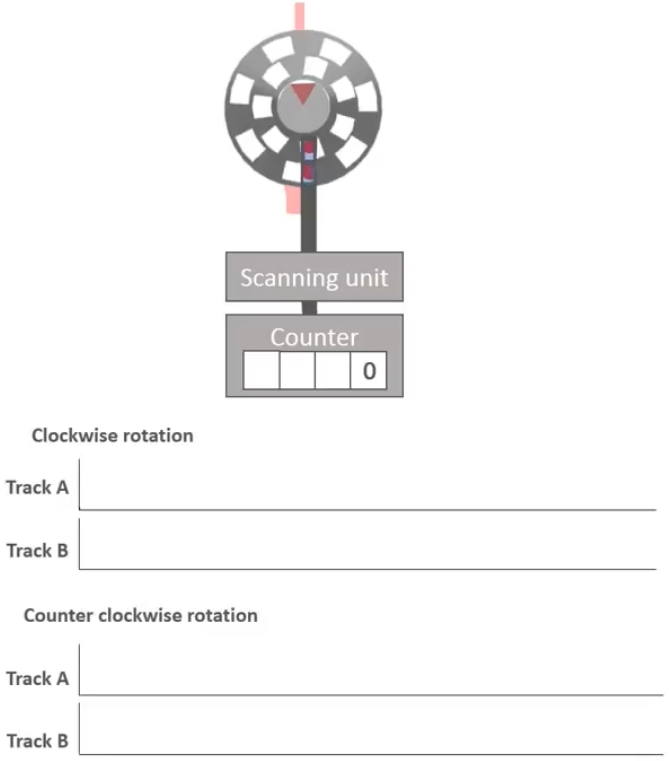
Source: youtube learnchannel

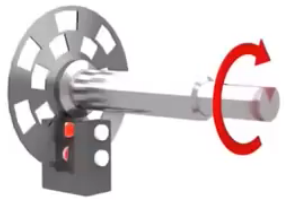
YouTube Learnchannel-TV

Odometry

Incremental Encoders

- In order to detect the **direction of rotation**, the encoder needs a disc with two tracks of gaps and two pairs of emitter/receiver diodes
- Based on the *order* of the pulses in the two tracks, the direction can be determined





Source: youtube learnchannel

YouTube Learnchannel-TV

Odometry

Absolout Encoders

- Absolute encoders use *coded discs* (where the gaps have patterns and are not equidistantly distributed)
- Based on that code, the angular state can be determined absolutely at any time

Source: youtube learnchannel

Odometry – Pros and Cons

Advantages

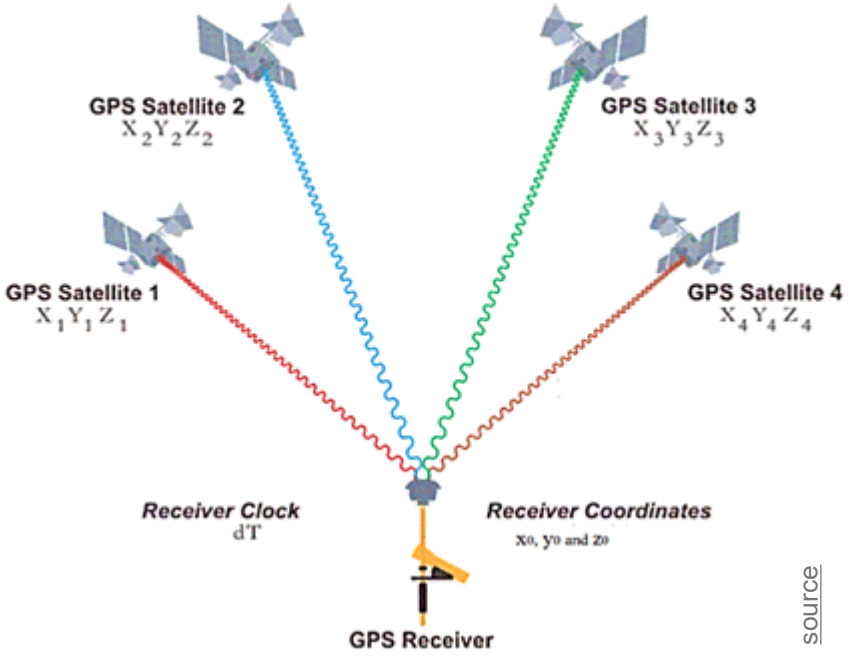
- operates independently (i.e. without needing external signals such as GPS or WiFi)
- is easy to implement and provides real-time position and yaw-angle estimation
- For example, if a robot has two wheels and both spin simultaneously forward → the travelled distance can be determined based on the number of wheel rotations

Disadvantages

- susceptible to cumulative errors (drift) over time: small inaccuracies in sensor data can lead to significant deviations in position and orientation estimates after a while
- can only provide relative positioning (based on known initial position)
- Changes in terrain or surface (e.g., slippery floors, uneven ground) can affect accuracy

GPS

- Global Positioning System (GPS) operates based on satellites that constantly transmit their current position and precise time using encoded radio signals
- GPS receivers can calculate their own position and speed by measuring the signal travel times
- With signals from three satellites, the receiver can compute its position in two dimensions (latitude and longitude), and with four satellites, it can also determine its altitude
- With GPS also the speed and direction of the receiver can be determined
- For details, see "Introduction to inertial sensors and navigation" in the QUT Robot Academy by Peter Corke



source

GPS – Pros and Cons

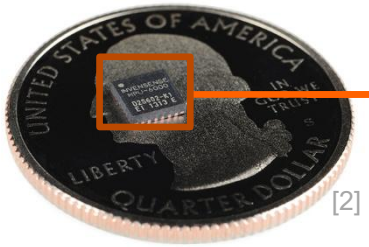
Advantages

- Offers global positioning (no knowledge regarding initial location required)
- Provides real-time data on location, speed, and direction
- GPS receivers available at low costs, and small, lightweight, and easily integrated

Disadvantages

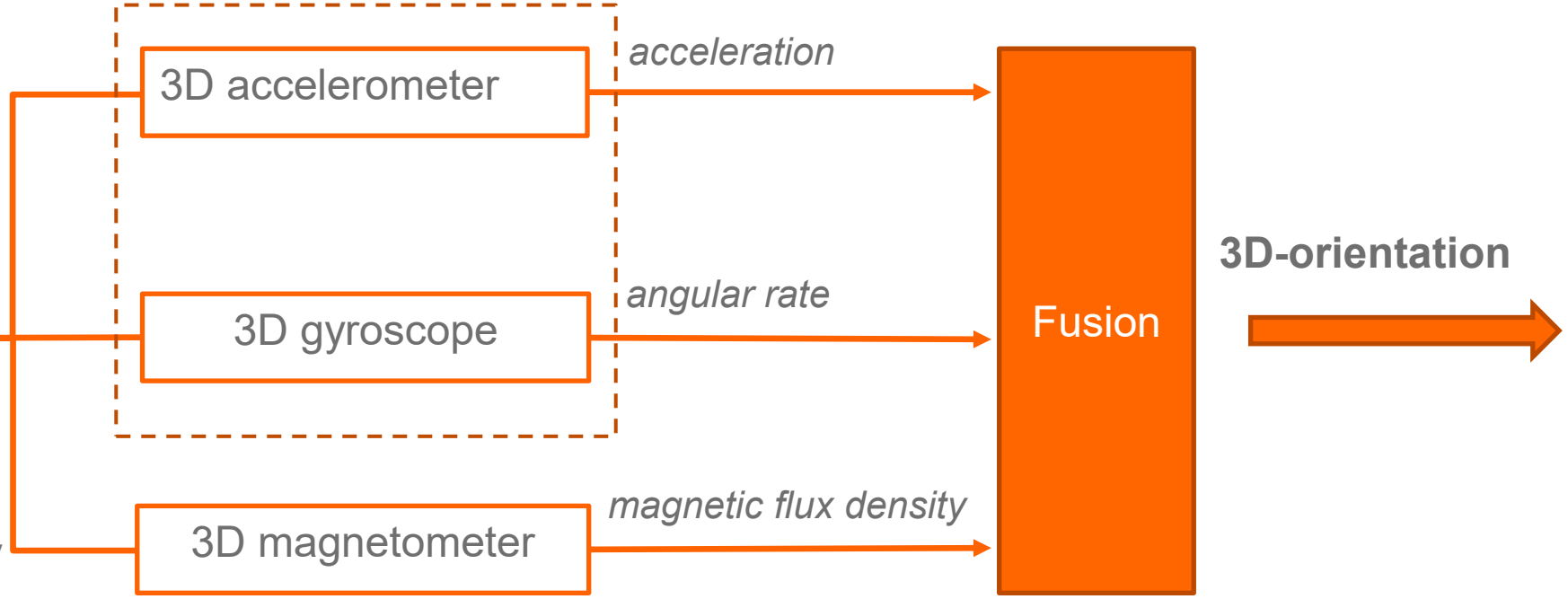
- in areas with poor visibility, such as indoors, in tunnels etc., GPS signals are blocked → no localization possible
- High power consumption (especially relevant for battery driven devices)
- Relatively low accuracy (>1m)

Inertial Sensors and Magnetometers



[2]

MARG sensor
Magnetic **A**ngular **R**ate **G**ravity



Inertial Measurement Unit - IMU
Combination of 3D accelerometer and 3D gyroscope (without 3D magnetometer)

Subject of the upcoming lectures!

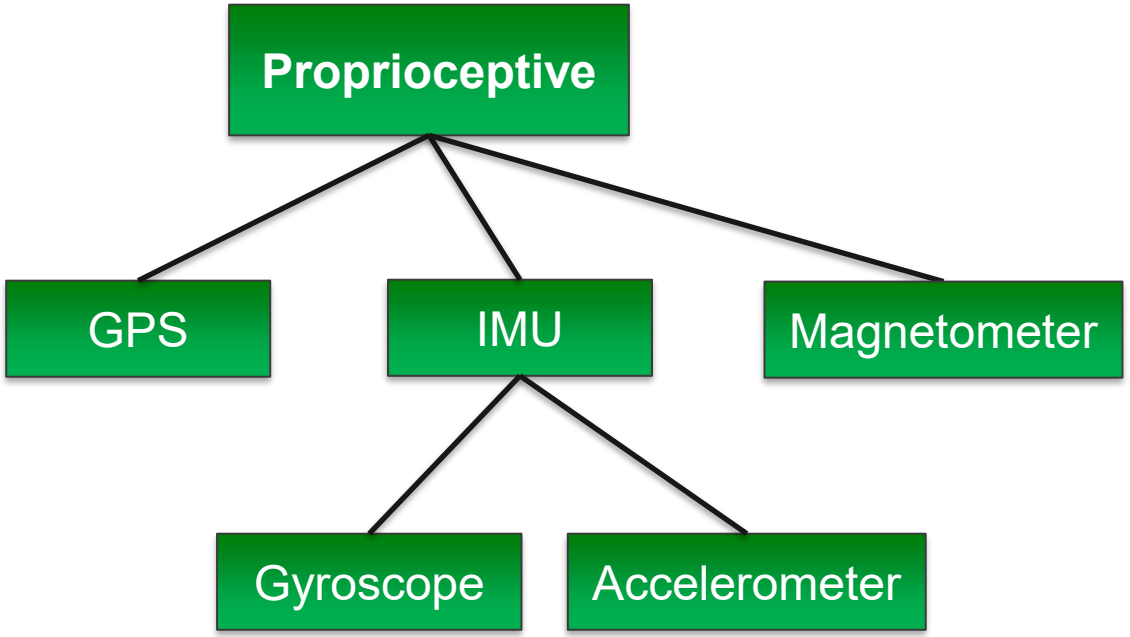
Categorization: Proprioceptive vs. Exteroceptive

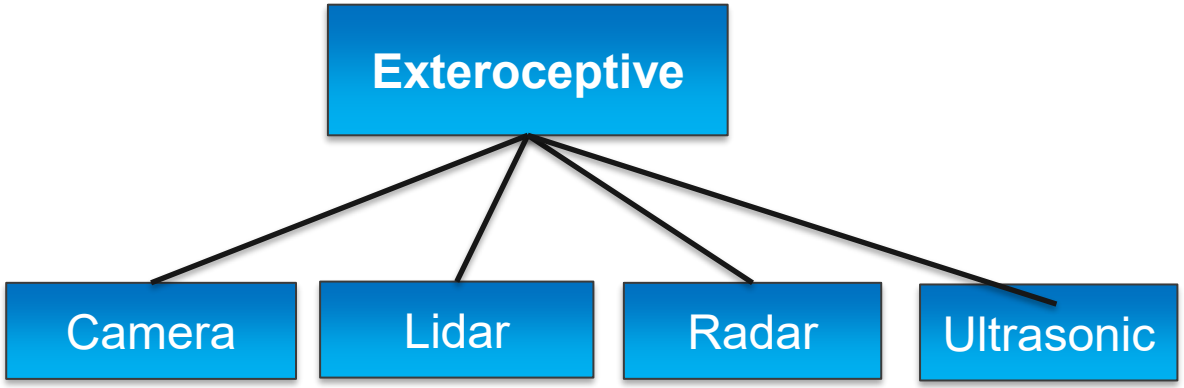
Proprioceptive sensors measure the internal state of the robot, focusing on its own position, motion, and orientation (latin word *proprius*, meaning "one's own" or "belonging to oneself“)

Exteroceptive sensors perceive information about the robot’s external environment, including objects, distances, and environmental conditions ("outside" or "external)

capere, meaning "to take" or "to capture."

Categorization: Proprioceptive vs. Exteroceptive



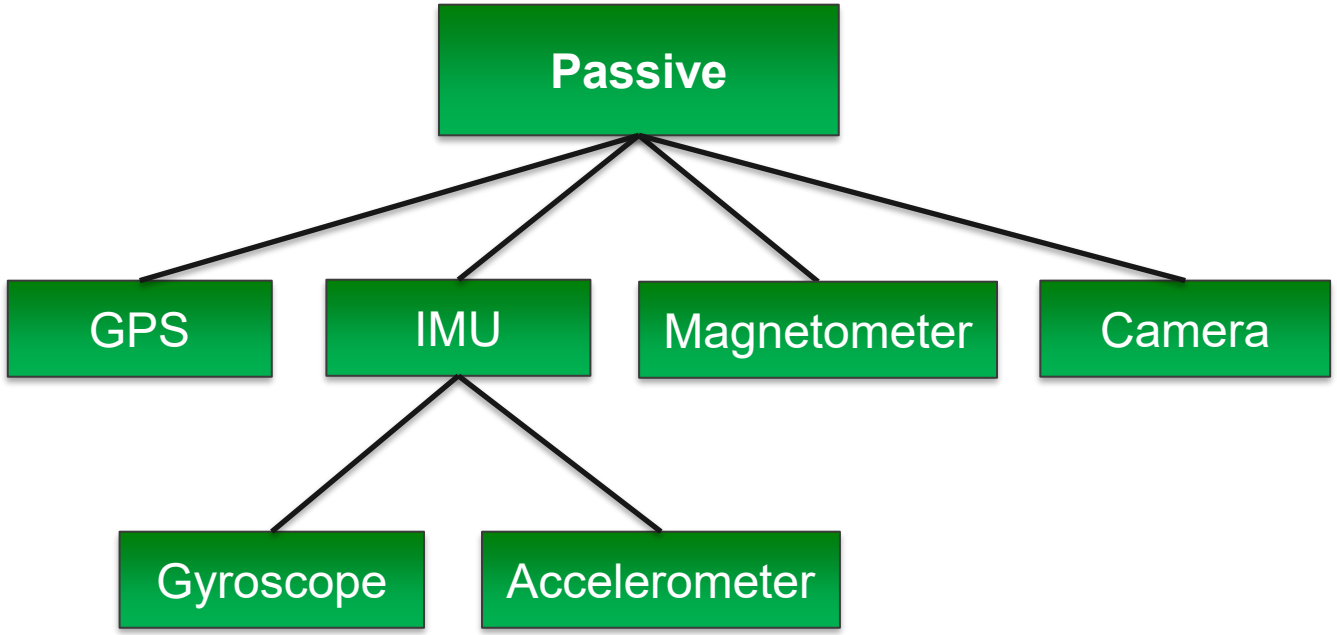


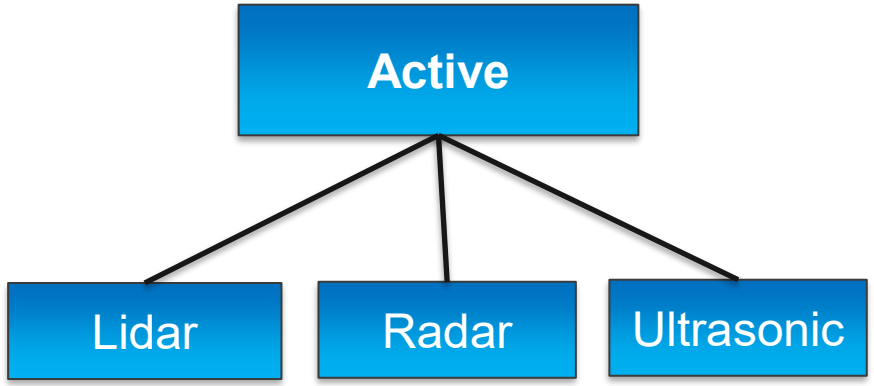
Categorization: Active vs. Passive

Active sensors emit signals and measure their return

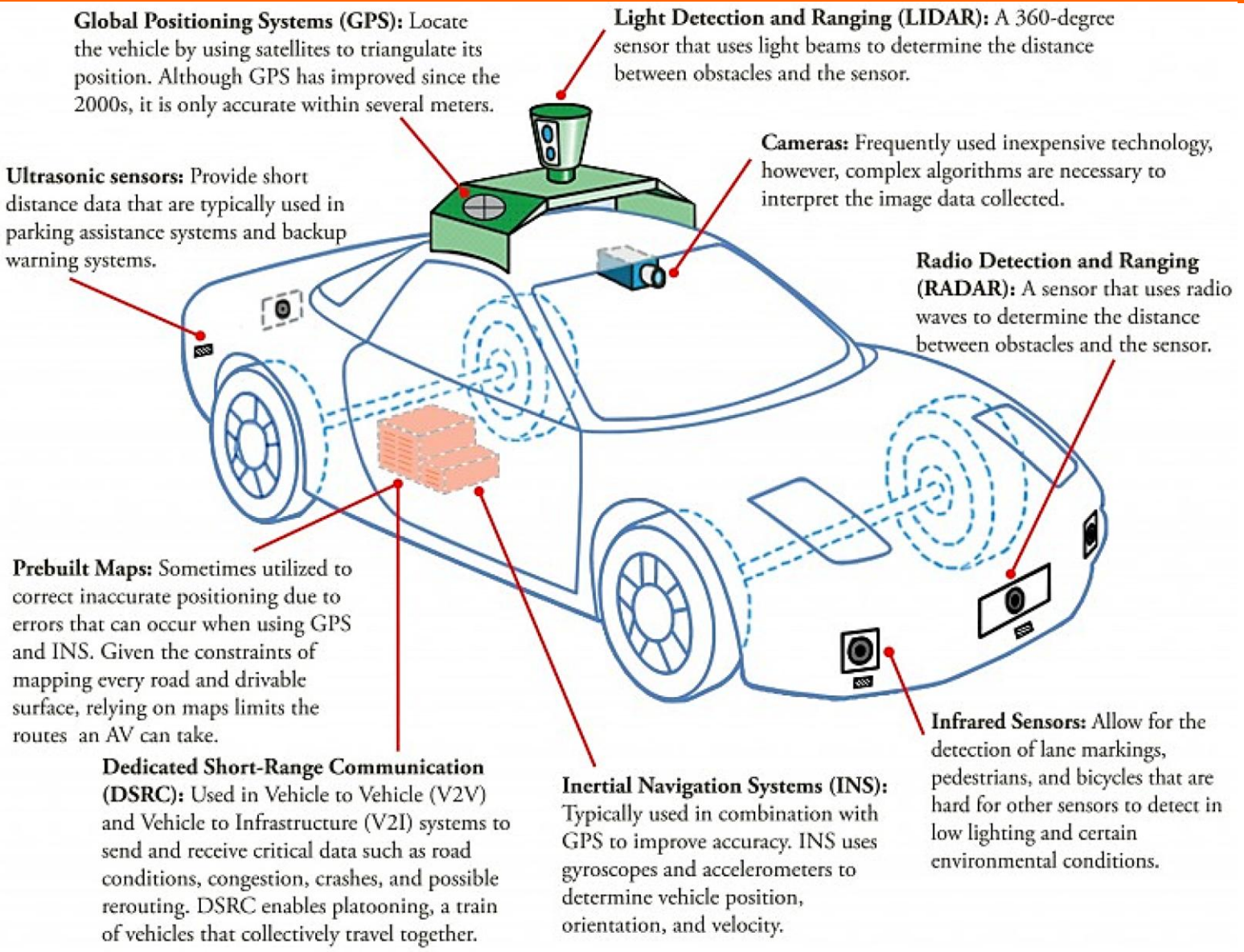
Passive sensors gather environmental information without emitting their own signals

Categorization: Passive vs. Active



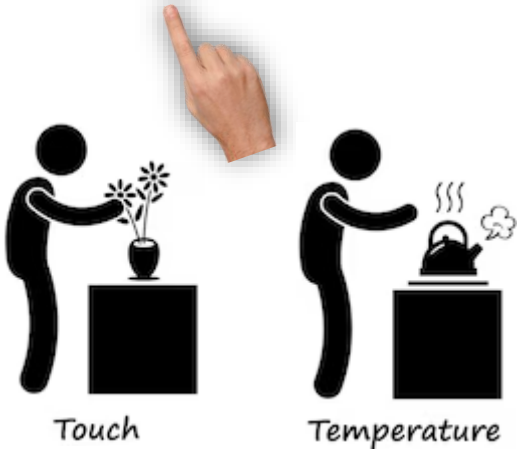


1 Perception in Mobile Robotics



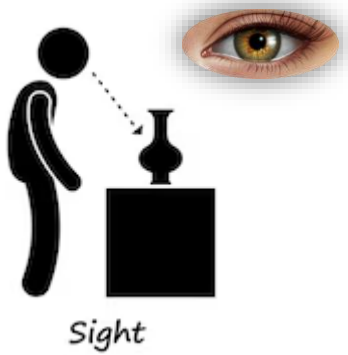
Source

- Mapping: human senses and robot senses



Bumpers, Ultrasonic sensors, Infrared sensors, Force sensors etc.

Temperature sensors



Camera, Lidar, Radar, Ultrasonic

Gas and chemical sensors

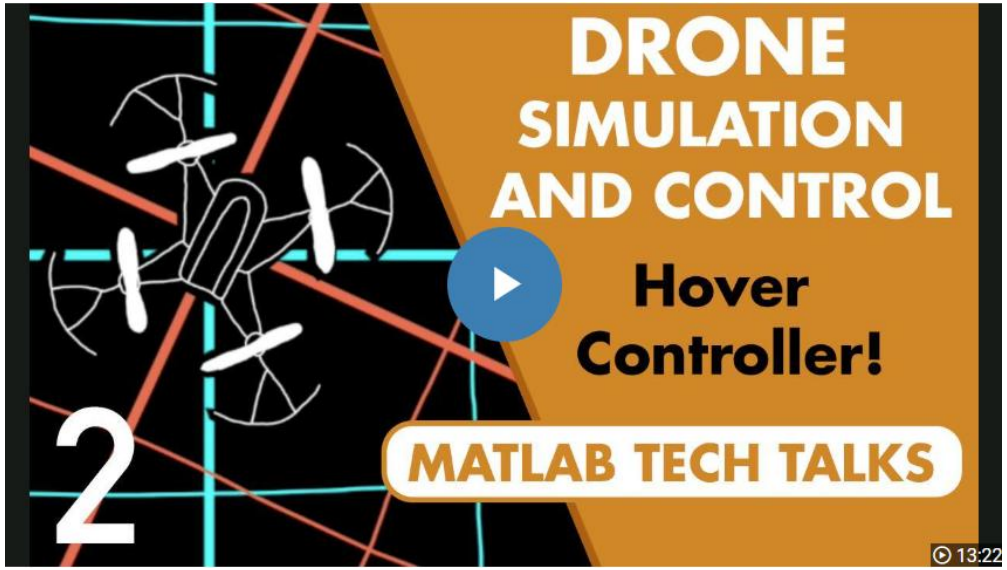
Microphones

IMU + Magnetometer

“Homework”:

Setting Up the Control Problem | Drone Simulation and Control, Part 1

https://de.mathworks.com/videos/drone-simulation-and-control-part-1-setting-up-the-control-problem-1539323440930.html



How Do You Get a Drone to Hover? | Drone Simulation and Control, Part 2

https://de.mathworks.com/videos/drone-simulation-and-control-part-2-how-do-you-get-a-drone-to-hover--1539323448303.html

References

[1] Die Sinne der Tiere, Stephan Frings, SpringerSpektrum (2021)

[2] Okajima, K., Nagaoka, S., Islam, M.R. et al. Development and testing of a surface roughness measurement device based on aerial ultrasonic reflections. Paddy Water Environ 18, 345–353 (2020). https://doi.org/10.1007/s10333-019-00786-3